

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Computer Science and Engineering) SEMESTER - 6 Summer 2025 ( Regular )**

**Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTCOC602 - COMPUTER NETWORKS**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 Which of the following best describes the difference between a protocol and a service in layered network design?
  - a. A protocol defines what a service does, while a service defines how it is implemented.
  - b. A protocol defines how services are implemented, whereas a service defines the interface between layers.
  - c. A protocol is implemented at the physical layer, whereas a service is only logical.
  - d. A service operates independently of any protocol
- 2 Consider a satellite communication link with a bandwidth of 10 Mbps and a round-trip delay of 500 ms. What is the bandwidth-delay product, and what does it signify in terms of performance?
  - a. 2.5 Mbits; amount of data that can be sent per second
  - b. 5 Mbits; total data in transit
  - c. 0.5 Mbits; latency
  - d. 10 Mbits; speed of transmission
- 3 Which of the following technologies is best suited for long-distance broadband wireless access in rural areas?
  - a. Wi-Fi (802.11)
  - b. Bluetooth (802.15.1)
  - c. WiMAX (802.16)
  - d. RFID
- 4 How does the token ring protocol prevent collisions in a network?
  - a. By detecting collisions after they occur
  - b. By assigning fixed time slots to each station
  - c. By using CSMA/CD
  - d. By allowing only the token holder to transmit
- 5 In the context of error detection, what is the role of the cyclic redundancy check (CRC)?
  - a. It corrects bit errors using redundancy.
  - b. It prevents packet loss.
  - c. It detects errors by performing polynomial division.
  - d. It provides flow control.
- 6 A frame uses a 4-bit checksum and is received as 10110010. The expected checksum is 1011. What can you conclude?
  - a. Error is corrected.
  - b. No error is detected
  - c. Error is detected
  - d. It is a broadcast frame.
- 7 Which routing algorithm uses periodic updates and can suffer from the count-to-infinity problem?
  - a. Link State Routing
  - b. Distance Vector Routing
  - c. Dijkstra's Algorithm
  - d. Path Vector Routing

- 8 If a network experiences variable delay and jitter, which congestion control technique is most appropriate?
- Token bucket
  - Leaky bucket
  - Flow-based admission control
  - Traffic shaping with priority scheduling
- 9 Which of the following statements best distinguishes between public key and private key cryptography?
- Both use symmetric encryption
  - Public key is faster than private key
  - Public key cryptography uses two keys, one for encryption and one for decryption
  - Private key cryptography requires certification authorities
- 10 Why is it important to implement firewalls at both the network and application levels?
- To increase system performance
  - To allow unrestricted access from the internet
  - To enforce multi-layered security policies
  - To simplify routing tables
- 11 Which HTTP method should be used when a client wants to update a specific resource without creating a new one?
- GET
  - POST
  - PUT
  - DELETE
- 12 A company's email system uses SMTP for sending emails and POP3 for receiving them. A user reports that sent emails are visible in the "Sent" folder but incoming messages are missing. Which layer or protocol is most likely misconfigured?
- HTTP
  - DNS
  - SMTP
  - POP3

Q2. Solve the following.

- Compare OSI and TCP/IP reference models with a suitable example. 6
- A network engineer is designing a new protocol. Evaluate the design issues they must consider at each layer of the OSI model. 6

Q3. Solve the following.

- Explain how Resilient Packet Ring (IEEE 802.17) combines the benefits of ring topology and packet switching. 6
- A college campus wants to deploy wireless connectivity over a large area. Which LAN technologies would you recommend and why? 6

Q4. Solve Any Two of the following.

- Explain the design issues of the Data Link Layer with suitable examples. 6
- Given a 7-bit message 1011001 and a 4-bit CRC generator polynomial 1101,
  - Compute the CRC code and form the transmitted message. 6
  - If the received data is 1011001010, detect whether an error has occurred during transmission using CRC. 6
- Explain in detail three different framing methods with the help of examples. 6

Q5. Solve Any Two of the following.

- Compare IPv4 and IPv6 in terms of address space, header structure, and security. 6

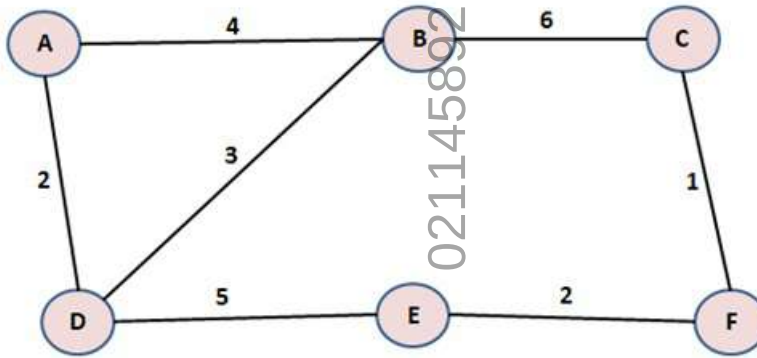
- B) Given the following network graph with nodes **A, B, C, D, E, and F** and their respective edge weights: 6

**Edge Weights:**

1. A-B = 4    2. A-D = 2    3. B-D = 3    4. B-C = 6  
5. C-F = 1    6. D-E = 5    7. E-F = 2

**Task:**

1. Apply Dijkstra's algorithm to compute the shortest path from source node **A** to all other nodes.
2. Show the final distance table and interpret the shortest path and cost from **A** to each node.



- C) Explain the basic working principle of the **Leaky Bucket** algorithm and how it enforces a constant output rate. 6

Q6. Solve Any Two of the following.

- A) Explain the role of DNS and describe how name resolution is performed. 6
- B) Explain why AES is considered more secure than DES. List at least three reasons with justification. 6
- C) Evaluate the role of digital certificates, public/private key cryptography, and firewalls in securing online banking transactions. 6

\*\*\* End \*\*\*